



POWER BI MAJOR CAPSTONE PROJECT
END TO END
POWER BI PROJECT GUIDE

PROJECT TITLE

TECH WORKFORCE SALARY &
FINANCIAL ANALYTICS DASHBOARD

PROJECT CATEGORY

HR Analytics
Salary Analytics
Financial Analytics
Workforce Analytics

INSTITUTE NAME

Infinity Code Nexus

TRAINER / GUIDE

Mr.Anand Kumar

CREATED BY

Oruganti Kiranmai

TOOLS USED

Power BI Desktop Power Query, DAX & Power BI
Service

SUBMISSION DATE

[15/06/2006]

Prepared as part of the Power BI Masterclass Program to demonstrate skills in Data Analysis, Data Modeling, DAX Development, Dashboard Design, UI/UX, Storytelling, and Business Intelligence Reporting.

PROJECT OVERVIEW

=====

PROJECT OBJECTIVE

Develop a comprehensive Power BI dashboard to analyze employee compensation, career growth, expenses, savings, skills, and workforce satisfaction within the technology industry.

Participants are expected to:

- ✓ Analyze salary structures
- ✓ Evaluate career growth
- ✓ Understand employee expenses
- ✓ Analyze savings patterns
- ✓ Assess job satisfaction
- ✓ Generate business insights

PROJECT GOAL

Transform workforce and financial data into actionable insights that help organizations understand employee compensation, retention, financial well-being, and workforce trends.

BUSINESS PROBLEM

Technology organizations need to understand how salaries, skills, experience, education, expenses, and job satisfaction influence employee performance and retention.

The objective is to identify:

- Which technology fields offer higher compensation?

Ans-Full stack developer

- How experience impacts salary?

Ans-More the experience higher the salary

- Do certifications increase income?

Ans-Yes

- How much do employees save annually?

Ans-On an average they save 16.53K

- What factors influence job satisfaction?

Ans-Promotion and overtime working hours

- Which employees are most likely to seek new opportunities?

Ans-One's who are not promoted

BUSINESS QUESTIONS

Participants must answer:

1. What is the average salary?

Ans-174.06k

2. Which technology field pays most?

Ans-Data science & ML

3. Which job roles earn the highest salaries?

Ans-Platform Engineer

4. How does experience affect salary?

Ans-Salary is high if experience is higher

5. Do certifications increase pay?

Ans-Yes

6. Which expenses consume the highest income?

Ans-Housing

7. Which employee groups have higher job satisfaction?

Ans-Full stack developers have higher job satisfaction

DATASET INFORMATION

=====

DATASET NAME : Tech Salary Dataset

DATASET DETAILS : Approximate Records 10,000+

Columns : 50

EXPECTED ANALYSIS AREAS

=====

Salary Analytics

Career Analytics

Expense Analytics

Savings Analytics

Skill Analytics

HR Analytics

Employee Satisfaction Analytics

PROJECT OBJECTIVES

OBJECTIVE-1

Salary Analysis : Analyze salary distribution, compensation structure, bonuses, and equity benefits.

OBJECTIVE-2

Career Growth Analysis : Evaluate the impact of experience, education, and promotions on salary.

OBJECTIVE-3

Expense Analysis : Analyze employee spending patterns and financial obligations.

OBJECTIVE-4

Savings Analysis : Evaluate annual savings and financial health indicators.

OBJECTIVE-5

Skills & Certification Analysis : Assess the relationship between skills, certifications, and earnings.

OBJECTIVE-6

Employee Satisfaction Analysis : Analyze job satisfaction, promotion history, and employee retention risk.

OBJECTIVE-7

Executive Reporting : Create dashboards suitable for HR Leaders, Finance Teams, Business Managers, and Executive Decision Makers.

EXPECTED OUTCOME

Participants should demonstrate:

- ✓ Data Understanding
- ✓ Data Analysis
- ✓ KPI Development
- ✓ Dashboard Design
- ✓ Storytelling
- ✓ Business Recommendations

DASHBOARD REQUIREMENTS

=====

Participants must create a minimum of 6 Dashboard Pages covering all major workforce, salary, and financial analytics areas.

1.EXECUTIVE OVERVIEW

=====

Purpose: Provide a high-level summary of workforce compensation, savings, expenses, and employee satisfaction.

Required KPIs

- ✓ Total Employees
- ✓ Average Annual Salary
- ✓ Average Annual Savings
- ✓ Average Bonus %
- ✓ Average Job Satisfaction Score
- ✓ Average Experience

Suggested Visuals

- ✓ Salary Distribution
- ✓ Top Technology Fields
- ✓ Employee Location Analysis
- ✓ Workforce Summary

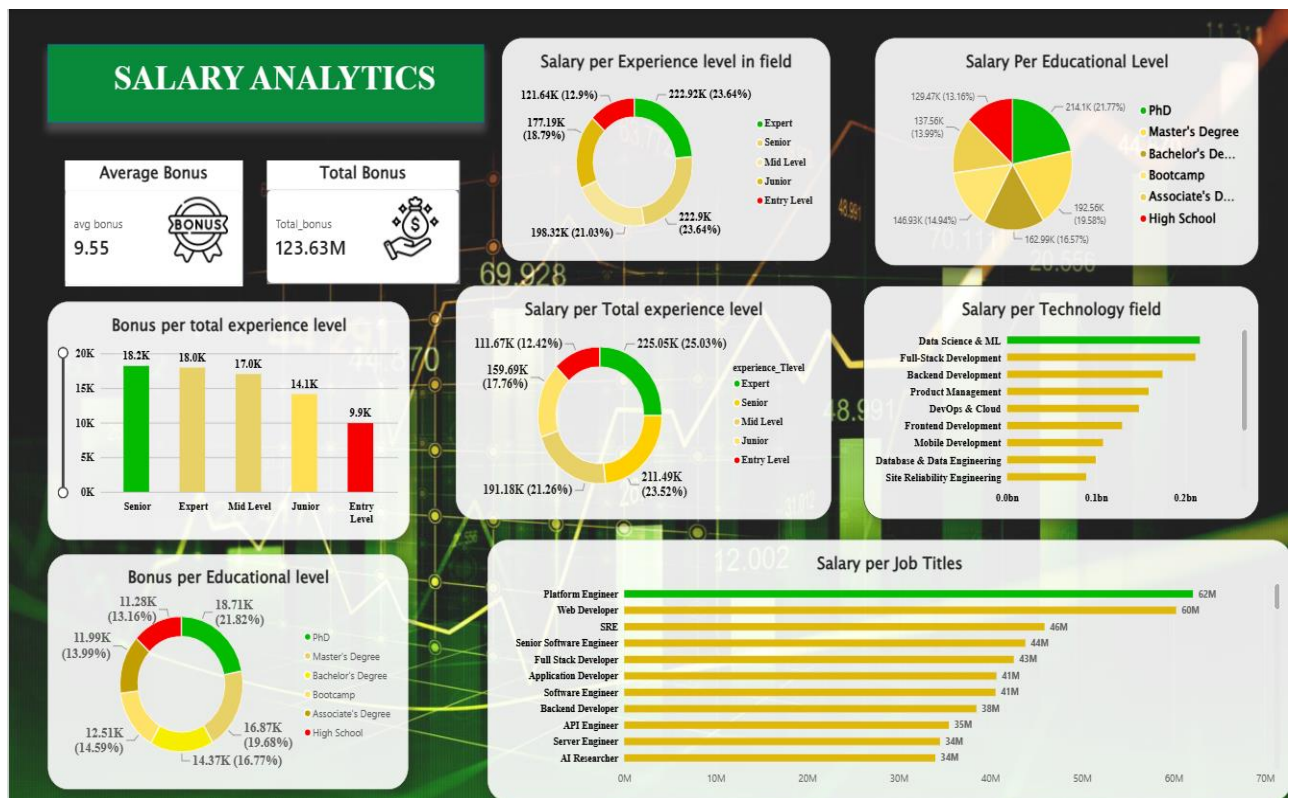


2.SALARY ANALYTICS

Purpose: Analyze compensation patterns across roles, skills, and technology domains.

Required Analysis :

- ✓ Salary by Technology Field
- ✓ Salary by Job Title
- ✓ Salary by Experience Level
- ✓ Salary by Education Level
- ✓ Bonus Analysis



3.CAREER GROWTH ANALYTICS

Purpose: Understand career progression and growth opportunities.

Required Analysis

- ✓ Experience vs Salary
- ✓ Promotion Analysis
- ✓ Previous Companies Analysis
- ✓ Education Impact on Salary
- ✓ Career Progression Insights

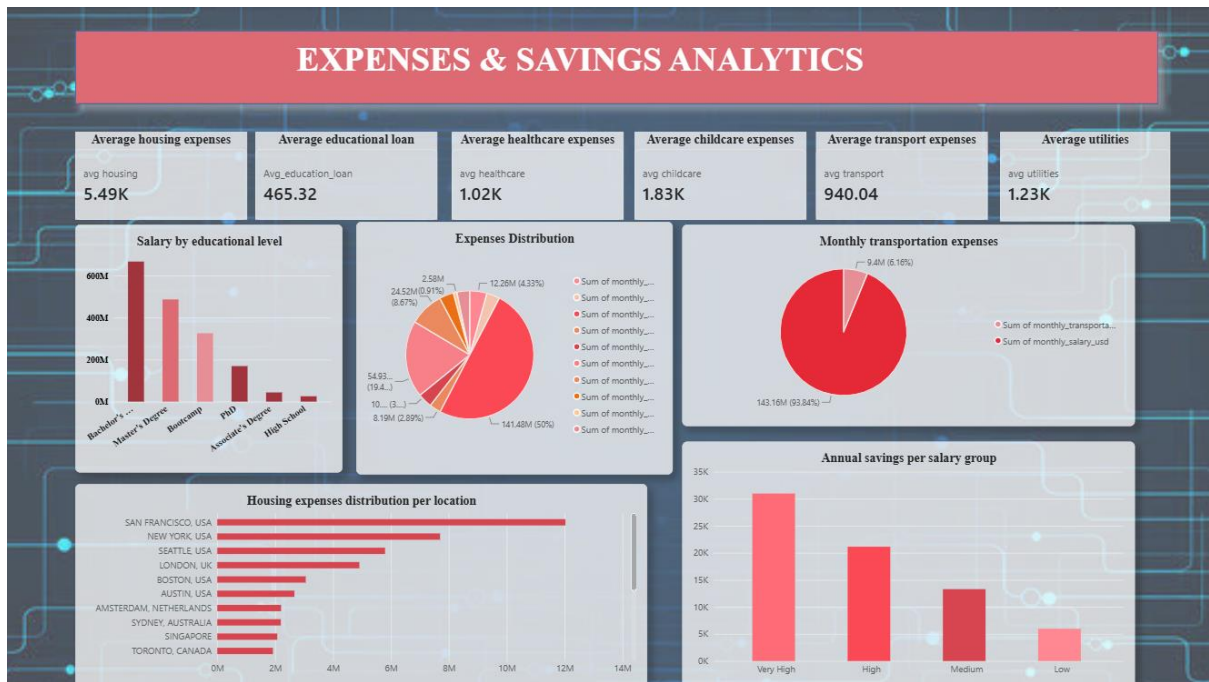


4.EXPENSES & SAVINGS ANALYTICS

Purpose: Analyze employee financial health and spending patterns.

Required Analysis

- ✓ Expense Breakdown
- ✓ Housing Expenses
- ✓ Transportation Expenses
- ✓ Annual Savings Analysis
- ✓ Savings by Salary Group



5.SKILLS & CERTIFICATIONS ANALYTICS

Purpose: Evaluate the impact of skills and certifications on earnings.

Required Analysis

- ✓ Salary by Primary Skill
- ✓ Certifications vs Salary
- ✓ Top Paying Skills
- ✓ Skill Distribution
- ✓ Technology Field Comparison

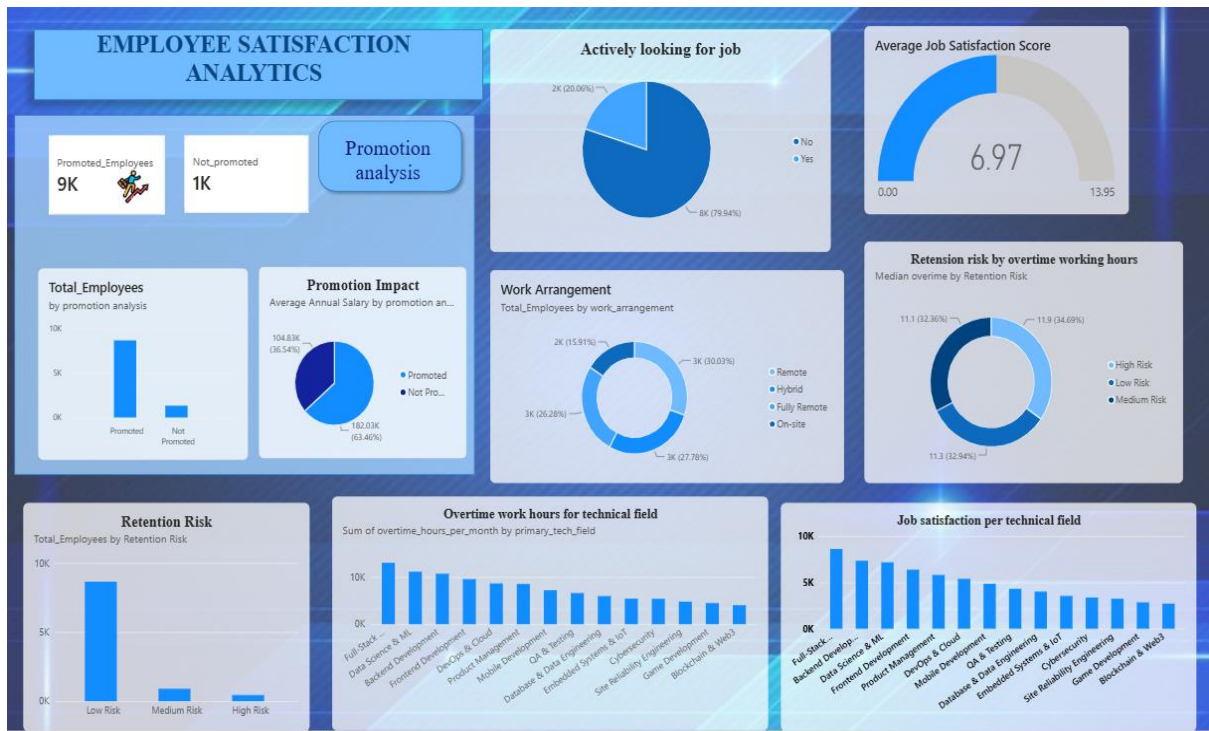
[illegible]

=====

Purpose: Analyze employee engagement, retention risk, and job satisfaction.

Required Analysis

- ✓ Job Satisfaction Analysis
- ✓ Promotion Impact
- ✓ Retention Risk Analysis
- ✓ Work Arrangement Analysis
- ✓ Actively Looking For Jobs Analysis



MANDATORY FEATURES

Participants must implement the following features.

DRILL DOWN

Examples:

Technology Field



Job Title



Employee

Location



Country

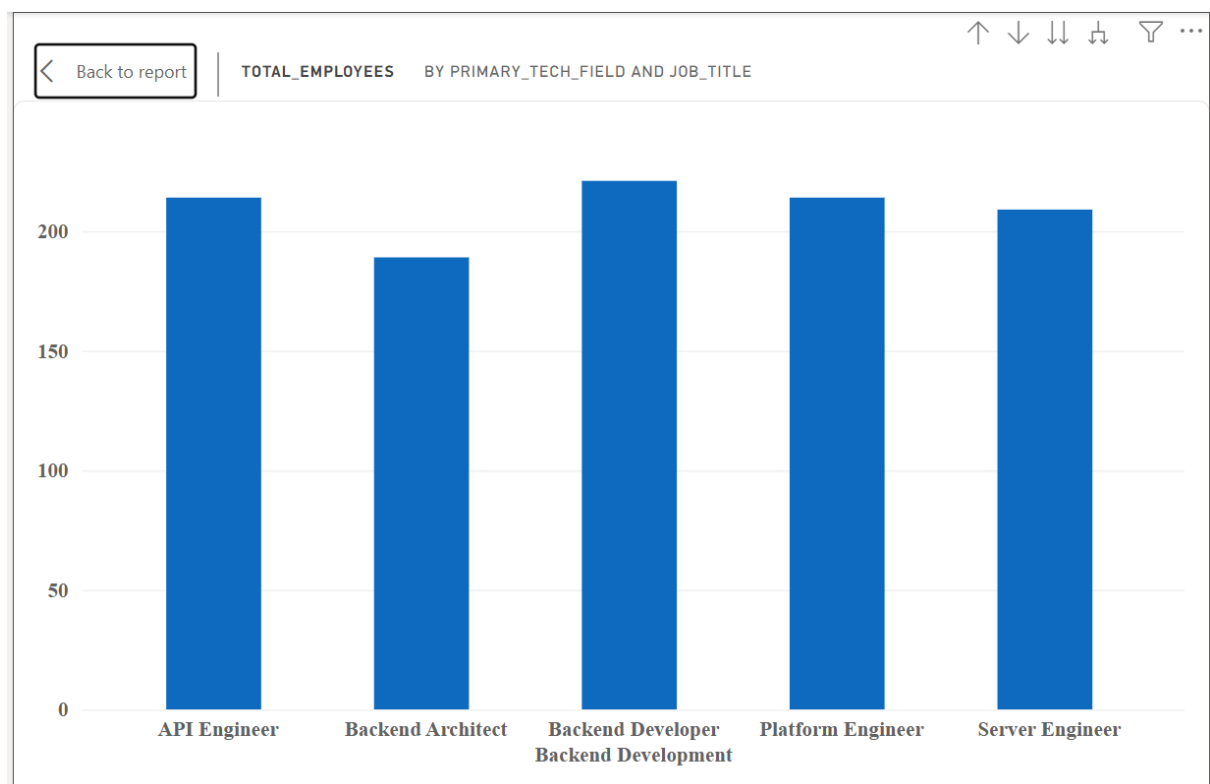
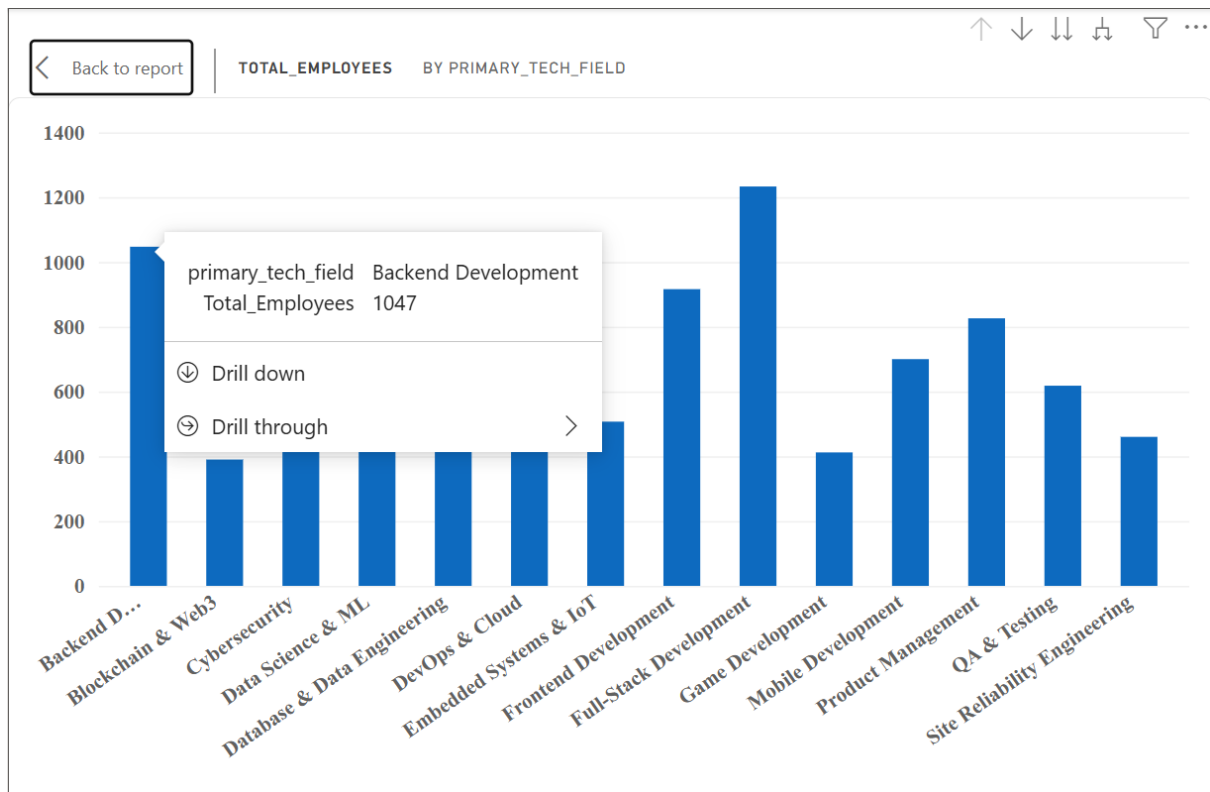


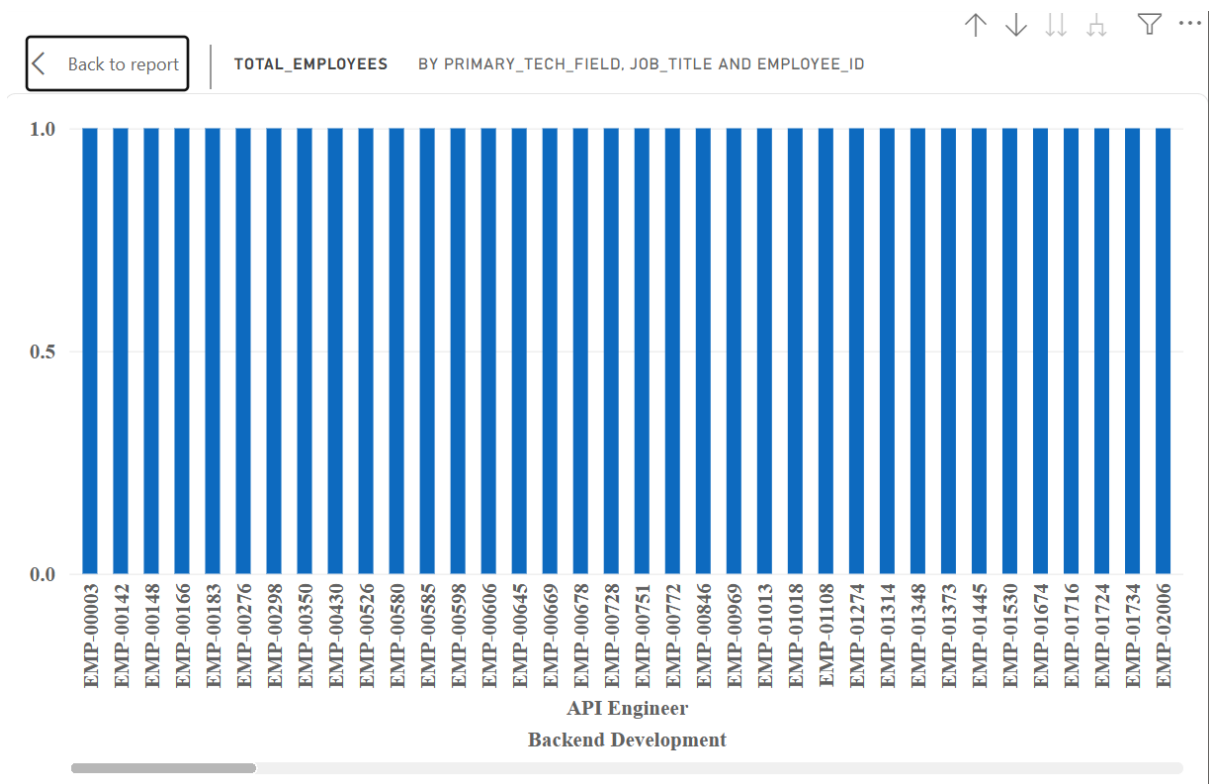
City

Experience Group



Employee Category





DRILL THROUGH

Examples:

Employee



Employee Profile Page

Technology Field

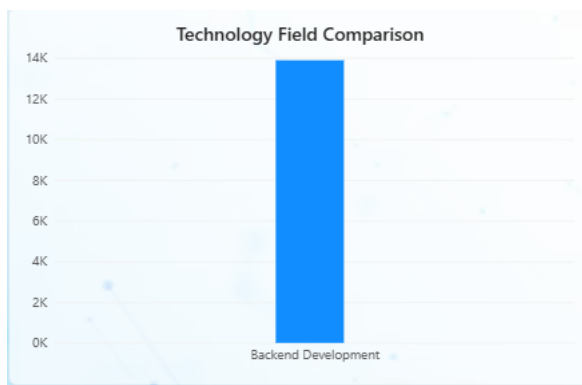
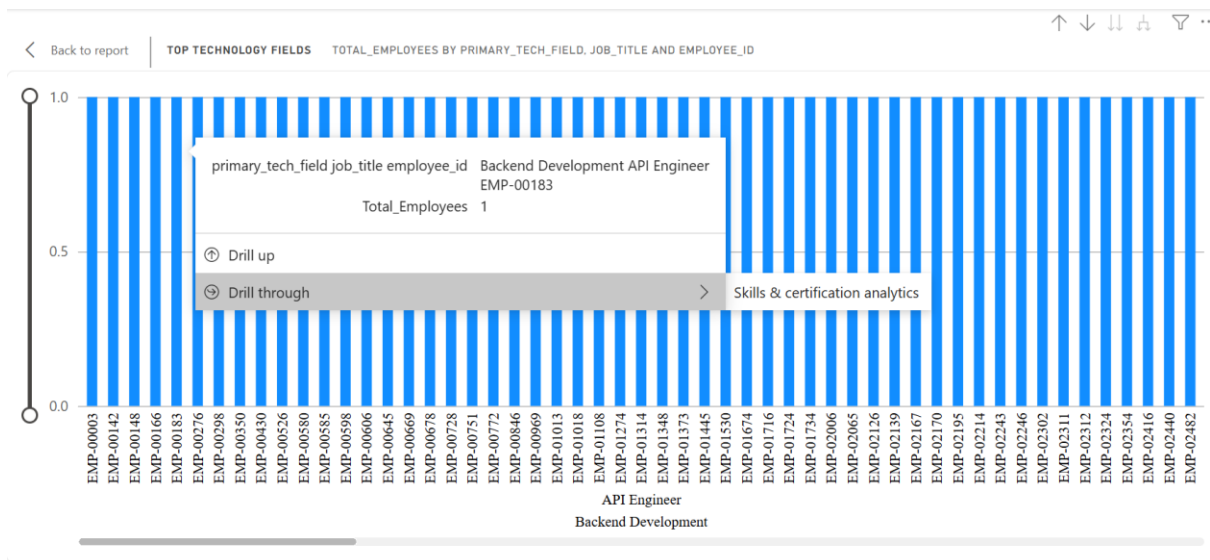


Technology Field Analysis Page

Job Title



Role Analysis Page



TOOLTIPS

Create custom tooltip pages for detailed employee insights.

CONDITIONAL FORMATTING

Apply DAX-based color logic for salary, savings, and job satisfaction metrics.

DYNAMIC TITLES

Titles must respond to slicer selections.

SLICERS

Required Slicers

✓ Technology Field

- ✓ Education Level
- ✓ Work Arrangement
- ✓ Gender
- ✓ Location

PROJECT DELIVERABLES

Participants must submit a complete Power BI project package consisting of the final PBIX file with all dashboard pages, KPIs, interactions, drill-through, drill-down, and reporting features.

Project documentation must include the project objective, business problem, dataset overview, KPIs created, insights generated, and business recommendations.

Screenshots of all dashboard pages must be included covering Executive Overview, Salary Analytics, Career Growth Analytics, Expenses & Savings Analytics, Skills & Certifications Analytics, and Employee Satisfaction Analytics dashboards.

A GitHub repository must be created containing the PBIX file, project documentation, screenshots, and a properly structured README file.

Participants must also deliver a professional presentation of 5–10 minutes explaining the analysis, insights, and recommendations.

JUDGING CRITERIA

- ✓ Accuracy
- ✓ Creativity
- ✓ Business Understanding
- ✓ Professional Design
- ✓ Insights Generated

SUBMISSION GUIDELINES

Participants must submit all project deliverables in an organized and professional manner. The final submission should include the PBIX file, project documentation, dashboard screenshots, GitHub repository link, and presentation materials.

All files should follow the prescribed folder structure and naming conventions to ensure smooth evaluation.

Participants are encouraged to publish their completed project on GitHub and showcase it on LinkedIn as part of their professional portfolio.

Late, incomplete, or improperly organized submissions may result in a reduction of evaluation marks.

PROJECT SUCCESS CRITERIA

“The dashboard should not only display workforce statistics.

It should answer HR and financial business questions.

It should generate meaningful insights about compensation, savings, career growth, and retention.

It should support business decision-making.

It should demonstrate professional dashboard design, storytelling, and analytical thinking.”